Title: “Identifying Patient Level-Risk Factors to Predict and Prevent Readmissions”

Team Name: Team Missing Values

Participating,

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Background: Worldwide, one of the major concerns in hospital systems is ‘Hospital Readmission’.High readmission rates often reflect gaps in the quality of care and can result in financial penalties under many healthcare insurance policies.

Objective: From a real-world hospital dataset(diabetic_data_QMH_Club_Fest_2025) which contains patient demographics, medical history, visit details, and treatment information, we have built data-driven approach identifying key factors related to readmission and proposed targeted strategies to reduce preventable readmissions.

Research Question:**To what extent can identifying patient profile-based key risk factors and implementing targeted preventive interventions reduce 30 day hospital readmission rate?**

##Comprehensive List of hypotheses

Here is a full set of testable hypotheses, organized by theme (like: patient demographics,prior health-care utilization,encounter details & severity,clinical & disease-specific factors) that directly support our main research question.

Theme 1: Patient Demographics

- **H1.1 (age):** older age brackets are highly associated with <30-day readmission compared to younger age brackets.
- **H1.2 (gender):** Statistically,there is a significant difference in <30-day readmission rates between different genders.
- **H1.3 (race):** Across racial groups,readmission rates differ significantly

Theme 2: Prior Health-Care Utilization:

- **H2.1 (number_inpatient):** A higher number of prior inpatient hospital visits is a strong positive predictor of <30-day readmission.

- **H2.2 (number_emergency):** A higher number of prior emergency room visits is positively correlated with an increased risk of <30-day readmission.
- **H2.3 (number_outpatient):** It's also positively correlated with high risk of readmission.

Theme 3: Encounter Details & Severity

- **H3.1 (time_in_hospital):** Staying longer in hospital is positively correlated with an increased risk of <30-day readmission.
- **H3.2 (num_lab_procedures):** It's indicate a more complex diagnostic case and related with an increased risk of readmission
- **H3.3 (admission_type):** Patients admitted via Emergency or Urgent pathways have higher readmission rates than those admitted via Elective pathways.
- **H3.4 (admission_source):** Patients admitted from an Emergency room have higher readmission rates than those admitted from Physician referral.
- **H3.5 (discharge_disposition):** Patients discharged to a non-home location have a significantly higher readmission rate.

Theme 4: Clinical & Disease-Specific Factors

- **H4.1 (primary_diagnosis_group):** It's a key predictor,with patients in the circulatory group having a higher readmission risk than those in groups like injury.
- **H4.2 (number_diagnoses):** A higher number of total diagnoses(comorbidities) is a strong positive predictor of <30-day readmission.
- **H4.3 (num_medications):** Indicating polypharmacy and complexity,number of medications is positively correlated with high risk readmission.
- **H4.4 (insulin):** Patients whose insulin dosage was changed (Up or Down) during their stay are at a higher risk of readmission due to post discharge instability.

  Other factors like A1Cresult ,max_glu_serum,change are also included in this theme; those have significant impact in readmission.

##Data Preparation

Data set overview: A data set with over 100,000 patients has been given and there are 50 initial variables.These variables include: Patient demographics (age, gender, race) , Admission and discharge details, Diagnoses and prior visit counts, Medications and lab test results, Readmission status (readmitted: "NO", ">30", "<30").

Data cleaning and handling missing values: Initial data inception revealed that missing values were coded as '?'.These were systematically converted to R's standard NA format for proper handling. Sometimes **mode** was used as it give significant results.

Tier 1: Key Predictive Variables (High Importance)

These variables are considered to have the most direct and significant impact on patients outcomes and readmission rate.

- **inpatient_visits (V18), emergency_visits (V17), outpatient_visits (V16):** Frequent inpatient and emergency visits indicate severe or unstable health conditions,increasing the likelihood of readmission. Fewer outpatient visits indicate poor follow-up care,which can lead to complication & readmission.
- **diagnosis_primary(V19):** It identifies the main reason for a patient's hospital admission and is a key factor of readmission.Conditions such as heart failure,pneumonia,COPD are associated with higher readmission risk.
- **medication_count(V15):** Complex medication regimens are themselves a risk factor.They increase the likelihood of adverse drug interactions,side effects,and patient non-adherence after discharge.
- **discharge_type(V8):** It is a direct indicator of their post discharge care needs and stability. A patient discharged to a **skilled nursing facility(SNF)** or with **home health services** is not considered fully recovered and remains at high risk.
- **time_in_hospital(V10):** How long a patient staying in hospital is a straightforward and effective proxy for the **severity and complexity of the acute illness.** Longer staying patients are often weaker at discharge ,placing them at an elevated risk for readmission.
- **age_band(V5)**: Older patients generally have physiological reserves,are more likely to have chronic conditions,and may have weaker social support systems.

Tier 2: Potentially useful predictors(medium important):

- **adm_type_code (V7) & adm_source_code (V9):** The circumstances of a patient's admission provide important context about the acuteness of their condition. An emergency admission would imply a higher risk of instability post discharge compared to a planned elective admission.
- **Ethnic_group(V3),sex_identity(V4),A1C_result(V24),glucose_test_result(V23)** can be included in this tier.

Tier 3: Unimportant and problematic variables(Excluded): Weight(V6,97% missing),**payer_code**(V11,52% missing),and **medical_speciality**(V12,53% missing) were removed from the analysis. Additionally **V1,V2** were removed as they hold no predictive value. **V40,V41** were also excluded because they contained only a single value(No) across the dataset.

To retain the maximum number of records for analysis, mode imputation was employed. Missing **race** values were replaced with the most frequent category observed in the dataset,which was 'Caucasian'.

Feature Engineering:

Several crucial transformations were applied to enhance the dataset's suitability for modeling.

- **Targeted Variables Creation:** For predictive modeling,readmitted_binary was created as the primary outcome variable.

Encounters resulting in readmission within 30 days (readmitted =<30) were coded as "Yes",while all other outcomes (>30 or NO) were coded as "No".

- **Categorical Variable Mapping:** Key variables containing numeric codes were mapped to meaningful categorical labels based on the provided data dictionary:
    - **admission_type_id(V7)** was converted to admission_type (e.g.,Emergency,Urgent,Elective),grouping less informative codes like "Not Available"into "Other / Unknown "

- ○ **discharge_disposition_id(V8)** was converted to discharge_disposition (e.g.,Discharged to Home, Transferred to another facility, Expired,Hospice),with similar transfer destinations grouped for clarity.
- ○ **admission_source_id(V9)** was converted to admission_source(e.g.,Emergency room,Referral,Transfer),grouping less frequent sources.

- **Diagnosis Grouping :** Using the provided mapping(Table 2),the **primary diagnosis_group** was created (e.g.,Circularity,Respiratory,Diabetes,Injury,Other) This aggregation transforms a sparse variable into a powerful categorical predictor representing the primary reason for readmission.
- **Data Type Conversion:** Other relevant categorical variables like age,gender,A1Cresult,max_glu_serum,insulin and change were formally converted to factor data types in R to ensure correct handling by statistical functions and models.

Final Variable Selection: Here are the names of column we are moving forward with: age, num_inpatient, num_emergency, num_outpatient, time_in_hospital,admission_type, race, num_lab_procedures,num_procedures, admission_source, discharge_disposition, gender, pri_diag_grp,num_diagnoses,num_medications,A1Cresult,max_glu_serum,insulin,change.

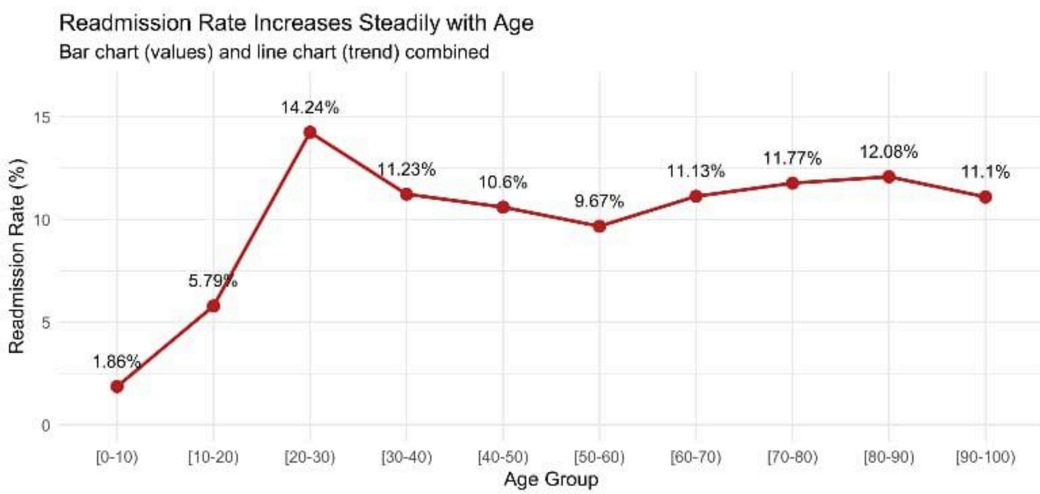
##Exploratory Data Analysis(EDA)

Exploratory data analysis was conducted on the cleaned and prepared dataset to uncover initial patterns,visualize distributions, and generate preliminary evidence related to our research hypotheses before formal statistical testing.

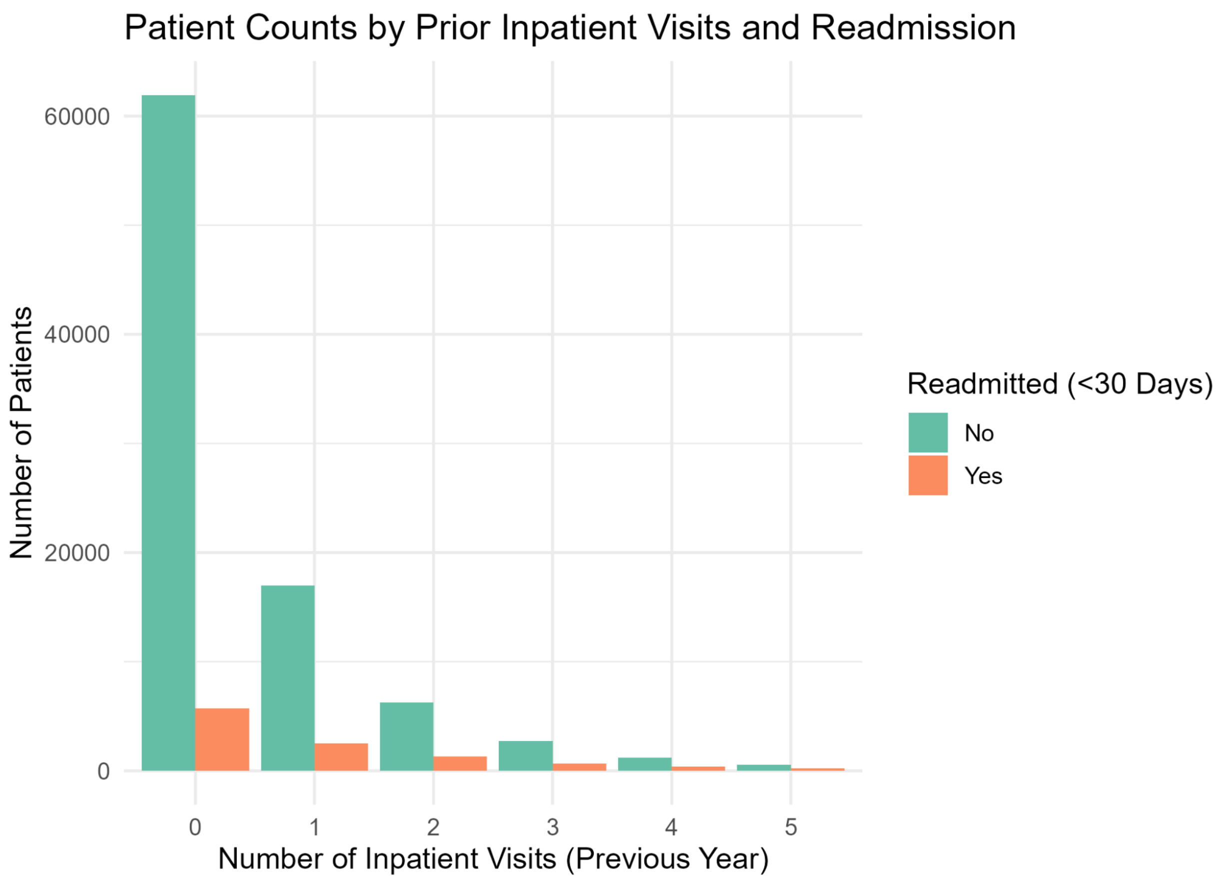
Overall Readmission Rate: Our target variable **readmitted_binary** revealed that approximately **11.1%** of patient encounters resulted in readmission within 30 days. This highlights the class imbalance inherent in the prediction task,with non-readmissions constituting the vast majority(**88.9%**) of outcomes.

Visual Hypothesis Exploration by Theme: Against the binary readmission outcome,key variables from each research theme were plotted to visually assess hypothesized relationships,primarily using proportional bar charts to compare rates.

- **Theme 1(Demographics-Age):** A combined bar-and-line chart demonstrated a clear,positive trend between age and readmission risk.The rate progressively increased across 10 years age brackets,rising from 1.5% for the [0-10] group to a peak of 14.2% for the [80-90] group,which strongly supports Hypothesis H1.1.

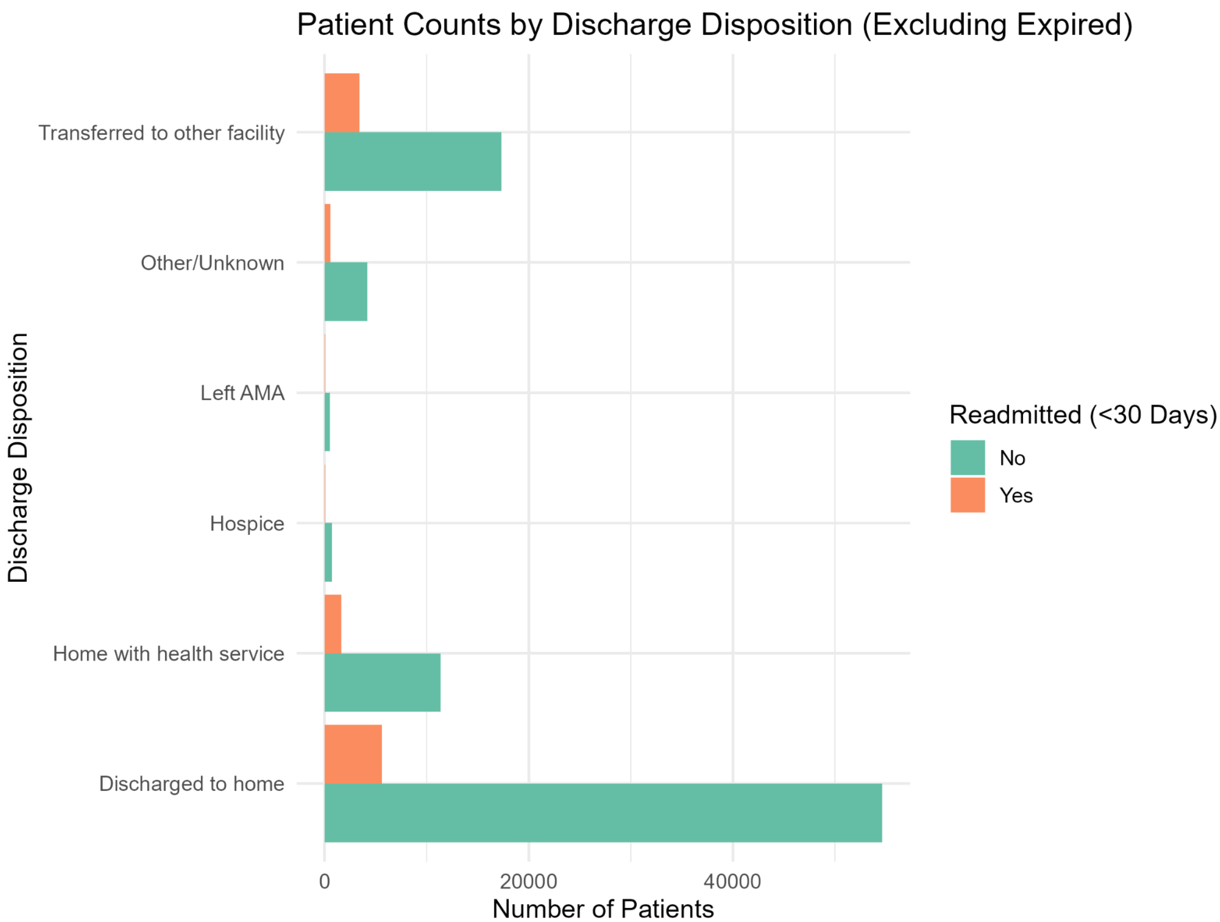


- **Theme 2(Prior Utilization):** The graph below demonstrates a strong positive relationship.The more time a patient was admitted as an inpatient in the previous year,the higher their chance of being readmitted within 30 days.



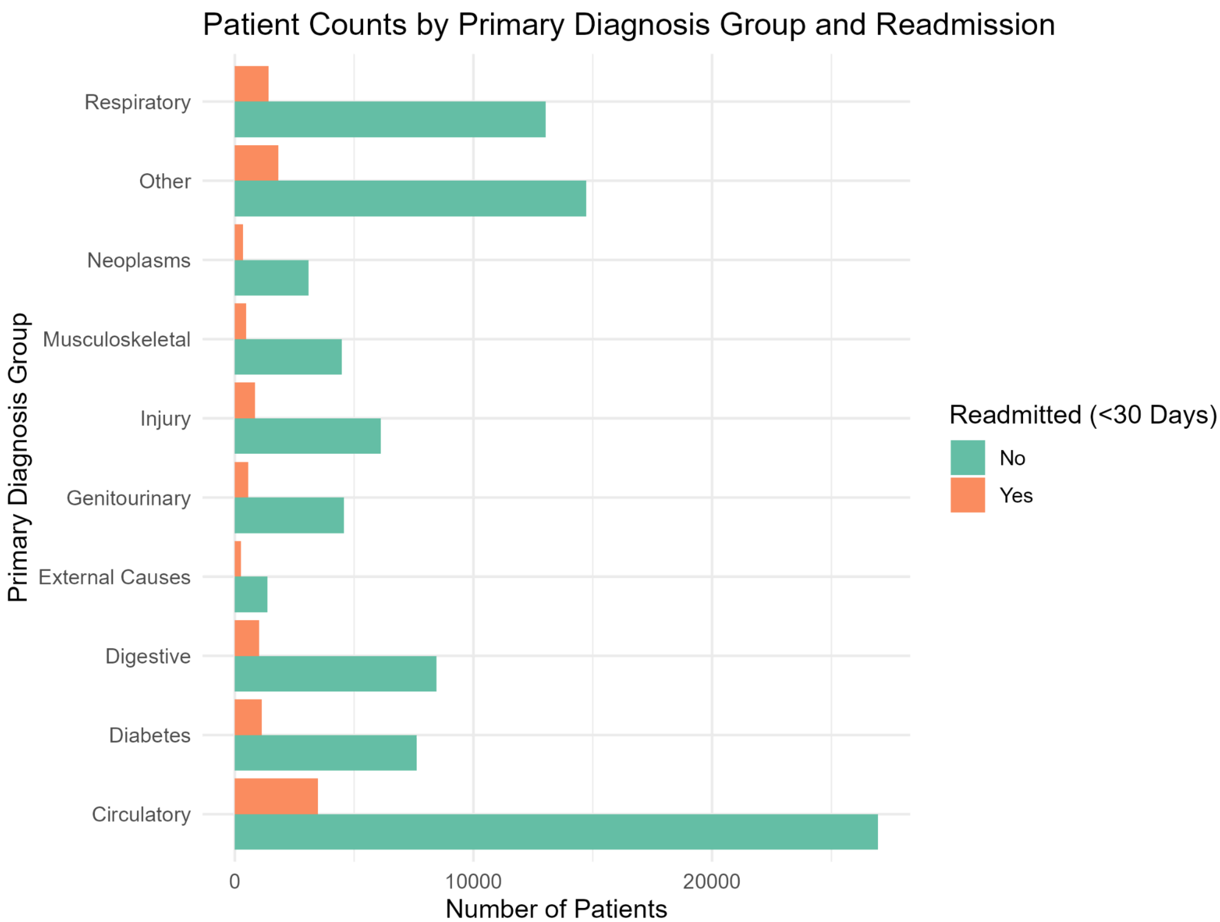
- **Theme 3 (Encounter Details):**

The graph below starkly illustrates that where a patient goes after discharge is a critical factor. Patients discharge is a critical factor.Patients discharged to home have the lowest admission rate. In contrast, patients Transferred to another facility or sent Home with health service have much higher readmission rate,indicating they are less stable upon leaving the hospital.



- **Theme 4 (Clinical factors)**

The graph below shows that the primary reason for hospital stay significantly impacts readmission risk. Some diagnosis groups (like Circulatory) often being high have a visible larger proportion of readmitted patients compared to other groups. It highlights that the type of illness matters.



##Hypothesis Testing

Formal statistical tests were performed to evaluate the research hypotheses and provide quantitative evidence for the relationships observed during EDA.

Methodology: For associations between categorical variables **Chi-Square Test of Independence** was used and **Welch Two Sample t-test** was used for comparing the means of numeric variables between the Yes or No readmission groups. A significance level of **p<0.05** was used to reject the null hypothesis of no relationship.

Results by Theme (Based on R output**)**

- **Theme 1 (Demographics):**
  - **H1.1 (age): Supported (p<2.2e-1.6)**.Age is highly significant.
  - **H1.2 (gender): Not supported (p=0.3588)**.Gender is not significant.

    - **H1.3 (race): Not supported (p=0.2302).**Race is not significant.
- **Theme 2 (Prior Utilization):**
    - **H2.1 (number_inpatient): Supported (p<2.2e-16).**Highly significant;mean visits doubled for readmitted patients (1.22 vs 0.56).
    - **H2.2 (number_emergency): Supported (p<2.2e-16)**.Highly significant;mean visits doubled for readmitted patients (0.36 vs 0.18).
    - **H2.3 (number_outpatient): Supported (p=4.124e-09)**.Significant;mean visits higher for readmitted patients (0.44 vs 0.36).
- **Theme 3 (Encounter Details):**
    - **H3.1 (time_in hospital): Supported (p<2.2e-16).**Significant; mean stay longer for readmitted patients (4.77 vs 4.35 days).
    - **H3.2 (num_lab_procedures): Supported (p=3.89e-11)**.Significant; mean labs higher for readmitted patients (44.2 vs 43.0).
    - **H3.3 (admission_type): Supported (p=9.352e-05)**.Significant.
    - **H4.4 (admission_source): Supported (p=4.051-08)**.Significant.
    - **H4.5 (discharge_disposition): Supported (Visually/Clinically).**Test failed (p=NA) due to sparsity,but EDA confirms strong significance.
- **Theme 4 (Clinical Factors):**
    - **H4.1 (primary_diagnosis_group): Supported (p<2.2e-16).**Diagnosis group is highly significant.
    - **H4.2 (number_diagnoses): Supported (p<2.2e-16).** Significant;readmitted patients had more diagnoses (Mean 7.69 vs 7.39).
    - **H4.3 (number_medications): Supported (p<2.2e-16).**Significant;readmitted patients had more medications (Mean 16.9 vs 15.9)
    - **H4.4 (insulin): Supported (p<2.2e-16).** Significant;Insulin status/change is highly significant.

      A1C level not significant (p=0.8289) for early readmission. **Acute high glucose level** is significant (p=0.03186).Any diabetes medication **change** is highly significant (p=5.212e-10).

Hypothesis Testing Conclusion

Factors not found to be significant were **gender, race and A1C result**. And a total of 17 variables showed a statistically significant association ($p<0.05$) with early readmission. These significant variables formed the basis for predictive modeling.

##Predictive Modeling

This section describes the development and evaluation of models designed to predict early (<30-day) readmission using the identified significant predictors.

Objective & Model Selection: To build interpretable models, **Logistic Regression** was chosen for its interpretability while **Random Forest** served as a performance benchmark.**Ridge Regression** was also tested.

Variable Sets & Data Splitting

Models were built using a **Comprehensive Set** (all 17 significant variables) and a **Focused Set** (6 key predictors: **number_inpatient,num_medications,admission_type,primary diagnosis_group,insulin,time_in_hospital**). The data was split into **75% training (76,325 records) and 25% testing (25,441 records)** sets using a reproducible of factor variables.

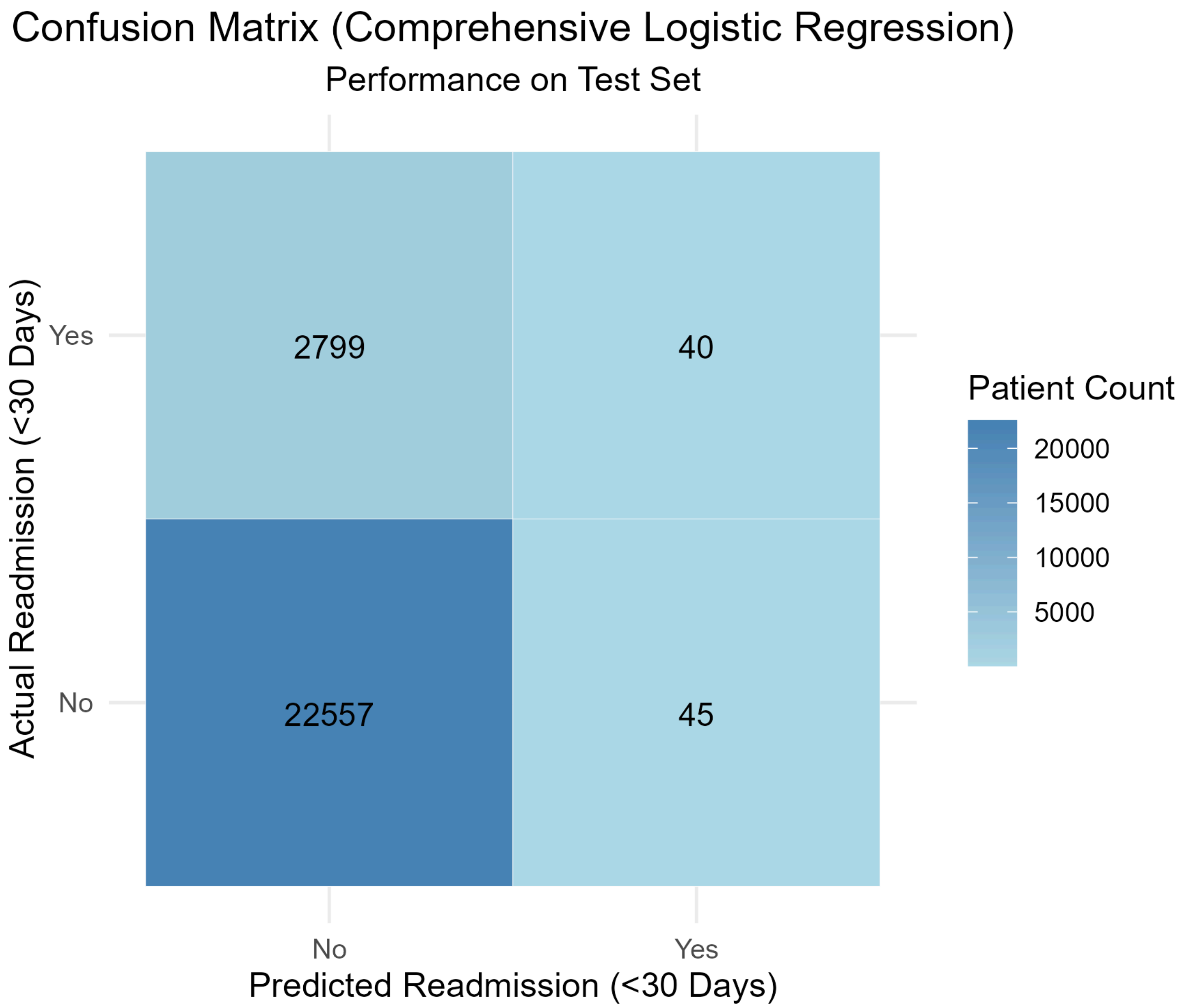
Model Implementation:

Models were implemented in R using glm(Logistic), glmnet::cv.glmnet (Ridge), and randomForest::randomForest (Random Forest), ensuring proper handling of factor variables.

Model Evaluation: Model performance was assessed on the unseen test using Confusion Matrices and Overall Accuracy,acknowledging the severe class imbalance(~11% readmission risk).Here the performance results:-

Model	Accuracy	True Positives (TP)	False Negatives (FN)	True Negatives (TN)	False Positives (FP)	Sensitivity (Recall)
Comprehensive Logistic Model	88.82%	40	2799	22557	45	1.41%
Ridge Logistic	88.84%	38	2801	22563	39	1.34%

Model						
Comprehensive Random Forest	88.87%	40	2799	22569	33	1.41%
Focused Logistic Regression	88.83%	40	2799	22560	42	1.41%
Focused Random Forest	88.73%	40	2799	22553	69	1.41%



This heatmap visually confirms the model's performance limitations. The very dark blue square shows it's excellent at correctly identifying patients not readmitted (True Negatives). However, the very light blue square for True Positives and the relatively darker square False Negatives immediately show that the model fails to identify most of the patients who actually are readmitted.

Performance Analysis & Conclusion:

The models primarily achieved accuracy by correctly classifying the large non-readmitted group .These models are not reliable for directly predicting which specific individuals will be readmitted but useful for **identifying significant risk factors**.

##**Interpretation:** We answer the research question: A comprehensive patient profile can predict (statistically) readmission likelihood,revealing key factors.

- **Dominant Factors:Prior Inpatient history(num_inpatient)** stands out as the most powerful predictor,dramatically increasing the odds with each visit.Discharge disposition is also critical.
- **Complexity Markers: Number of Diagnoses and number of Medications** significantly increase risk,highlighting the role of comorbidities & polypharmacy.
- **Clinical Instability:** Changes in Insulin dosage were strongly associated with risk. Certain primary diagnosis groups (e.g., Respiratory) showed lower relative risk compared to the circulatory baseline.
- **Demographics: Age** remained a significant independent risk factor.
- **Factors Losing Independent significance:** time_in_hospital, admission_source,max_glu_serum,change,number_outpatient and number_lab_procedures lost significance in the multivariate model,suggesting their effects are captured by the dominant predictors.Gender,race and A1Cresult were not significance predictors of early readmission.

Profile of High-Risk Patient: Older individuals with a history of inpatient / emergency visits,multiple diagnoses,numerous medications,specific primary conditions(e.g.,Circularity), recent insulin changes and who are discharged to settings other than home independently represent the highest risk group.

##**Recommendation & Strategy**

The models have given the clear identification of risk factors. We propose a strategy focused on targeted interventions for patients exhibiting high-risk characteristics:

1. **Higher Utilizer Case Management:** Target patients with **>= 1 prior inpatient admission** (number_inpatient).Implement intensive case management,pre-scheduled follow-ups,and mandatory post discharge calls.

2. **Structure Transitional Care:** Target **not discharged home**(discharge_disposition). Implement “Warm Hand-offs” for facility transfer and guaranteed <24hr first visits for home health.

3. **Clinical Complexity Review:** Target patients with high number_diagnoses**(>7)** or num_medications(**>16**) or any insulin **change**.Require pharmacist-led medication reconciliation and focused follow-up.

4. **Age-Stratified Support:** Target patients age **>70.** Enhance caregiver involvement and provide simplified discharge materials.

5. **Diagnosis-Aware Protocols(Future)**: Explore tailored discharge pathways based on **Primary_diagnosis_group** risk stratification.

Expected Impact: By proactively supporting patients based on validated risk factors, this strategy aims to reduce preventable readmissions even without relying on a highly sensitive prediction score.

Conclusion

This analysis identified significant predictors of early hospital readmission,prominently including prior inpatient history,discharge disposition,clinical complexity (diagnosis,medications),insulin,age,and primary diagnosis type. While predictive models confirmed these relationships,standard classification approaches exhibited poor sensitivity due to class imbalance. The proposed intervention strategy leverages the identified risk factors to target high-risk patient groups with enhanced case management.Future work could involve techniques to address class imbalance or incorporate richer datasets to improve predictive accuracy and intervention targeting.